

PRANAV JAIN

University of Southern California

Email: pranavj@usc.edu

[Website](#) , [Scholar](#)

RESEARCH INTERESTS

Geometry Processing, Physical Simulations, Computer Graphics, Numerical Analysis of Partial Differential Equations, Discrete Differential Geometry

EDUCATION

University of Southern California (USC)
Doctor of Philosophy (PhD), Computer Science

August 2023 - Present
Los Angeles, USA

New York University (NYU)
Master of Science (MS), Scientific Computing

September 2021 - May 2023
New York, USA

Indraprastha Institute of Information & Technology Delhi (IIITD)
Bachelor of Technology with Honors (B-Tech Hons),
Computer Science and Applied Mathematics

August 2016 - August 2020
New Delhi, India

PUBLICATIONS

Pranav Jain, Navami Kairanda, Peter Yichen Chen, & Oded Stein. (2026). PINNsur: Physics-Informed Neural Networks for PDEs on Curved Surfaces. [\[link\]](#)

Pranav Jain, Ziyin Qu, Peter Yichen Chen, and Oded Stein. 2024. Neural Monte Carlo Fluid Simulation. In ACM SIGGRAPH 2024 Conference Papers (SIGGRAPH '24). [\[link\]](#)

Liam Martin, **Pranav Jain**, Zachary Ferguson, Torkan Gholamalizadeh, Faezeh Moshfeghifar, Kenny Erleben, Daniele Panozzo, Steven Abramowitch, Teseo Schneider. A systematic comparison between FEBio and PolyFEM for biomechanical systems. In Computer Methods and Programs in Biomedicine, 2023. [\[link\]](#)

Zachary Ferguson, **Pranav Jain**, Denis Zorin, Teseo Schneider, and Daniele Panozzo. High-Order Incremental Potential Contact for Elastodynamic Simulation on Curved Meshes. In ACM SIGGRAPH 2023 Conference Proceedings (SIGGRAPH '23). [\[link\]](#)

Pranav Jain, Munawar Hasan, Donghoon Chang. Spy based analysis of selfish mining attack on multi-stage blockchain. In Cryptology ePrint Archive, 2019. [\[link\]](#)

RESEARCH EXPERIENCE

Google Summer of Code
Contributor

May 2026 - August 2026
Remote

Advisor: Prof. Martin Skrodzki, Dr. Andreas Fabri

- Implemented the algorithm proposed in [Lipschütz et al. \(2024\). Manifold Meshing of Point Clouds with Guaranteed Smallest Edge Length](#) for mesh reconstruction from scratch and integrated it into the [Computational Geometry Algorithms \(CGAL\)](#) library

Adobe Research
Research Intern

May 2025 - August 2025
Seattle, USA

Advisor: Dr. Timothy Langlois, Dr. Danny Kaufman, Dr. Jui-hsien Wang

- Researched on how to make generative-AI models more physics accurate by limiting hallucinations for video generation

University of Southern California

PhD Student

August 2023 - Present

Los Angeles, USA

Advisor: Prof. Oded Stein, Prof. Krishna Garikipati

- Researching on empirically proving convergence rates of Poisson problem using Physics Informed Neural Networks (PINN) with respect to the size of the architecture
- Researched and published on a discretization-free fluid simulation method using combination of Monte-Carlo and neural fields

New York University

Research Assistant

September 2021 - May 2023

New York, USA

Advisor: Prof. Daniele Panozzo, Prof. Denis Zorin

- Researched and compared the simulation abilities of PolyFEM and FEBio in terms of accuracy, time and user inputs
- Introduced a high-order finite element formulation for elastodynamic simulation with contact handling

nTopology

Software Engineer Intern in Geometry Team

June 2022 - August 2022

New York, USA

Advisor: Suraj Musuvathy, Ranjeeth Mahankali

- Designed and implemented a method to efficiently stitch different CAD models to form one unified CAD structure

Freie Universität Berlin [\[link\]](#)

Research Intern

September 2020 - August 2021

(Virtual) Berlin, Germany

Advisor: Prof. Konrad Polthier, Dr. Sunil Kumar Yadav

- Researched on a robust and parameter-tuning free method to denoise a point cloud

Fields Undergraduate Summer Research Programme 2020 [\[link\]](#)

Research Intern

July 2020 - August 2020

(Virtual) Toronto, Canada

Advisor: Prof. Thomas Uchida

- Researched on the problem of inverse mechanism synthesis – given a path, design a mechanism that traces the path

Indraprastha Institute of Information & Technology Delhi [\[link\]](#)

Research Assistant

August 2018 - August 2020

New Delhi, India

Advisor: Prof. Kaushik Kalyanaraman

- Developed DECAT, a software framework for discretizations of the objects and operators of exterior calculus

Indraprastha Institute of Information & Technology Delhi [\[link\]](#)

Undergraduate Thesis

August 2018 - August 2020

New Delhi, India

Advisor: Prof. Donghoon Chang

- Proved the selfish mining attack strategy on the Bitcoin blockchain for multiple mining pools
- Designed and proved an extended attack for multi-stage blockchain

ACADEMIC COMMUNITY WORK

Tertiary reviewer for SIGGRAPH 2026

February 2026

Tertiary reviewer for SIGGRAPH Asia 2025 and Pacific Graphics (PG) 2025

July 2025

Volunteered as a teaching assistant for the Poisson Reconstruction Project at [Summer Geometry Initiative \(SGI\) 2024](#)

July 2024 - August 2024

TEACHING EXPERIENCE

University of Southern California

Teaching Assistant: CSCI 532 - Innovation for Defense Applications	Spring 2026
Teaching Assistant: CSCI 103 - Introduction to Programming	Fall 2025
Teaching Assistant: CSCI 580 - 3D Graphics and Rendering	Spring 2025
Teaching Assistant: CSCI 596 - Scientific Computing and Visualization	Fall 2024
Teaching Assistant: CSCI 104 - Data Structures and Object Oriented Design	Spring 2024

New York University

Grader: MATH 263.3 - Applied Partial Differential Equations	Spring 2023
Grader: MATH 252 - Numerical Analysis	Spring 2023
Grader: MATH 263 - Partial Differential Equations	Fall 2021